CADDN: A Content-Aware Downsampling-Based Detection Method for Small Objects in Remote Sensing Images

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***Abstract*— A key issue of existing deep-learning-based object detection methods in remote sensing images is that they often struggle to differentiate the background and small object regions due to multilevel downsampling operations therein. Downsampling operations help extract high-level semantic features but result in excessive loss of spatial features of small objects. In this article, we propose a new small object detector using multispectral remote sensing images, named content-aware downsampling-based detection network (CADDN), where we newly design a content-aware downsampling-based module (CADM). Unlike conventional downsampling operations that apply uniform downsampling parameters across the entire feature map, CADM adaptively assigns higher weights to feature elements that are critical for distinguishing objects from the background, and this assignment is guided by the contextual awareness of object locations during the downsampling process. Experiments based on multispectral remote sensing images with small ships and vehicles demonstrate that CADM can accurately identify and preserve the locations of important object-related features, and CADDN correspondingly achieves superior small object detection performance than state-of-the-art methods.**

***Index Terms*— Content-aware downsampling, multispectral remote sensing, small object detection.**

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I. Introduction

DETECTION of objects in remote sensing images [2], [3], [4], [5], [6], [7], [8] plays an important role in both civilian and military realms, providing essential support in maritime rescue, urban planning, traffic navigation, and defense deployment [9], [10], [11].

In recent years, object detection in remote sensing images using deep neural networks (DNNs) [3], [4], [5], [6], [7], [8] has significantly advanced, owing to its strong capability for autonomous feature extraction from data and superior detection performance in complex backgrounds compared to handcrafted feature-based methods [12]. One of the major challenges in remote sensing image detection is the presence of small objects, which we define based on their pixel size. Objects occupying only a few pixels in an image pose significant challenges for detection. For example, in Sentinel-2 images with a spatial resolution of 10 m, small objects may occupy fewer than 32 × 32 pixels, corresponding to a real-world area of less than 320 × 320 m^2. Fig. 1 presents typical scenes of ship and vehicle detection in remote sensing images. Objects in these images have extremely limited appearance information, which increases the difficulty of learning discriminative feature patterns during training DNNs and to accurately identify objects during inference.

Detecting small objects in remote sensing images has attracted much attention from researchers. DNN-based small object detection methods can be broadly categorized into improved training strategy-based methods and advanced feature engineering-based methods. Notably, DNN-based methods often show degraded small object detection performance with limited training data [13] or when training strategies are sensitive to positional bias [14]. Improved training strategy-based methods refer to improvements in various aspects of training DNNs, such as data augmentation and improved loss function design. Data augmentation is an intuitive and effective way to improve the performance of small object detection, such as synthesizing small object samples through generative adversarial networks (GANs) to improve the diversity of the training set [13] and improve data quality through super-resolution [15], [16]. The normalized Wasserstein distance (NWD) metric [14], which is robust to

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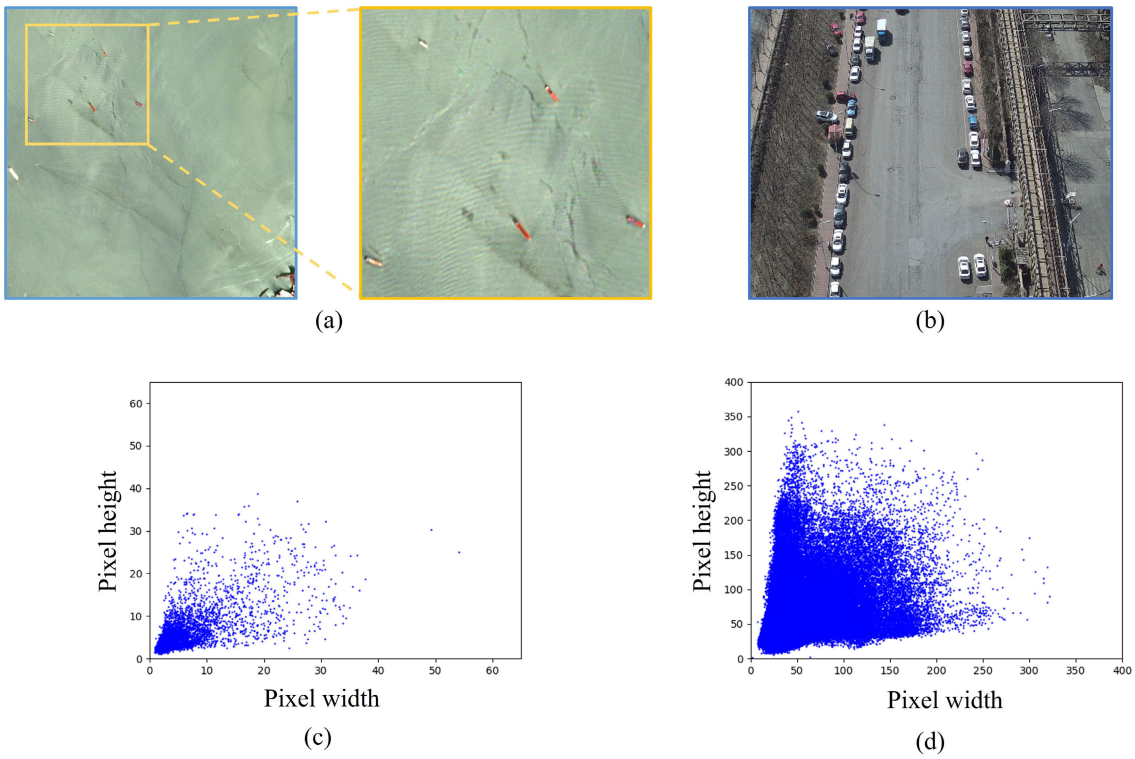


Fig. 1. (a) Representative scenes of MTSD. (b) Representative scenes of DroneVehicle. (c) Statistical results of width and height of objects in MTSD. (c) Statistical results of width and height of objects in DroneVehicle. The size of the objects is measured in pixel.

position deviation, is introduced to design the loss function and label assignment strategy that is more suitable for training small target data. Considering that feature engineering is the foundation of object detection, enhancing the feature representation capability of small objects is another effective detection strategy. Advanced feature engineering-based methods include contextual feature learning methods [17], [18] that utilize the contextual information of objects, feature-level super-resolution methods [19], [20], and multispectral feature mining-based methods [9], [20], [21], [22].

Note that existing DNN-based object detection methods often employ spatial downsampling operations to reduce the dimensionality of the feature space. Typical downsampling operations include average pooling, max pooling, and convolution with a stride greater than 1. These downsampling methods help decrease memory consumption and capture high-level semantic features with an extended receptive field. However, these operations might hinder the extraction of discriminative features for small objects, making it challenging to effectively preserve and aggregate the important features, even leading to the issue of losing small objects [23]. Fig. 2(a) illustrates the detection results of the baseline model, which employs conventional downsampling via convolution with the stride of 2, resulting in the omission of several small objects. The baseline model is based on the add-fusion multispectral detector described in [9], using infrared (IR) and RGB data as the default multispectral inputs. Fig. 2(b) shows the feature maps before and after the classic downsampling operations within the baseline model. As indicated by the red circles, features associated with small objects suffer a significant loss during the downsampling process, leading to missed detections as shown in Fig. 2(a). Therefore, despite the progress made by existing methods in small object detection, the potential loss of discriminative features during downsampling remains overlooked.

In this article, we propose a content-aware downsampling-based detection network (CADDN), which incorporates a novel content-aware downsampling module (CADM) to mitigate the loss of small object features during downsampling, thereby enhancing small object detection performance. CADM enhances the preservation and aggregation of small-object-relevant information by encoding the content of input features and predicting the spatial locations of key information. As shown in Fig. 2(c), CADDN successfully detects the small objects missed by the baseline model. As depicted in Fig. 2(d), our proposed CADM can better preserve and aggregate small-object-related features during the downsampling process in comparison with conventional downsampling operations.

The contributions of this article are summarized as follows. To address the challenge of small object detection, we propose a novel downsampling module named CADM, which replaces traditional downsampling methods and serves as a key component of the advanced small object detector CADDN. CADM effectively preserves and aggregates critical features based on content awareness. Besides, we design the parallel backbone to leverage additional spectral and spatial information from multispectral data to improve detection performance.

The rest of this article is organized as follows. Section II reviews the related work. Section III describes the proposed small object detection framework, including the overall

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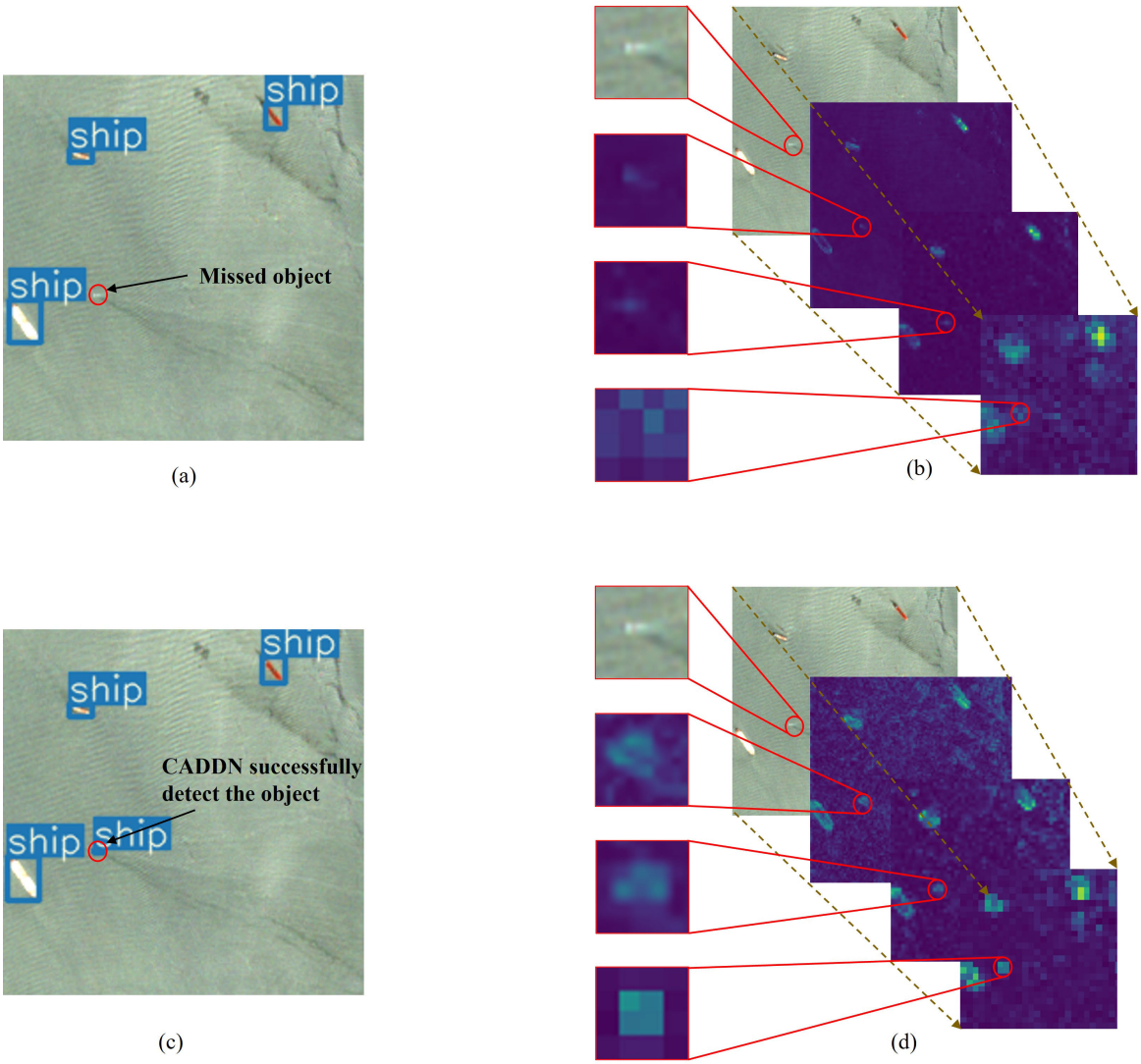


Fig. 2. Detection results and corresponding visualizations of feature maps for both the baseline model and the proposed CADDN. (a) Detection results of the baseline. (b) Visualization of the feature maps for the baseline model. (c) Detection results of CADDN. (d) Visualization of the feature map for CADDN.

architecture, the details of CADM, and the loss function. Section IV presents experimental results and analysis, including dataset descriptions, experimental setup, and performance evaluation. Section V concludes our work and discusses directions for future research.

II. Related Work

In this section, we review previous work closely related to this article. First, we introduce small object detection methods relevant to CADM from two perspectives: improved training strategy-based methods, and advanced feature engineering-based methods. Then, we summarize recently proposed downsampling techniques. Finally, we clarify the relationship between our proposed CADM and existing approaches.

A. Small Object Detection Methods

Training effective small object detectors is a challenging yet crucial task due to several inherent difficulties, such as training bias caused by the small proportion of small objects in datasets and the high sensitivity of the loss function to positional deviations owing to the tiny object size. One effective approach for small object detection is improved training strategy-based methods, such as data augmentation. Data-augmentation-based methods aim to diversify training samples, reduce training bias, and enhance the robustness of detectors, thereby enabling better detection of small objects across diverse scenarios. Common data augmentation techniques include resizing, rotation, flipping, and cropping. Kisantal et al. [24] utilize a copy-paste strategy to increase the occurrence of small objects, improving detection accuracy. Bosquet et al. [13] employ GANs to generate realistic small objects from larger ones, further enriching dataset diversity. In [15], super-resolution is used as a preprocessing step to enhance image resolution, thus improving the detectability of small objects. However, data augmentation methods often require time-consuming manual annotations or meticulously designed algorithms for synthetic data generation. Singh and Davis [25] propose selective training for multiscale objects within specific size ranges, improving

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adaptability across different object scales. Wang et al. [14] address the sensitivity of intersection over union (IoU)-based label assignment for small objects by modeling predicted bounding boxes as 2-D Gaussian distributions and introducing a novel label assignment strategy based on NWD, which enhances detection performance.

Advanced feature engineering-based methods focus on designing better feature representations tailored for small object detection. Since contextual information is often critical for recognizing small objects, Gidaris and Komodakis [18] propose the multiregion convolutional neural network (MRCNN), which fuses object features with their surrounding context to improve detection. Li et al. [17] propose the large selective kernel network (LSKNet), which leverages prior context information in remote sensing scenes by dynamically adjusting the receptive field to compensate for missing fine-grained features. Zhang et al. [20] introduce a super-resolution branch to reconstruct high-resolution images from feature maps, enabling better detail capture. Chen et al. [19] develop the degraded reconstruction enhancement network (DRENet), which leverages degradation reconstruction to make the network focus more on small objects. Wang et al. [26] propose the content-aware upsampling algorithm (CARAFE), which generates upsampled feature maps with large receptive fields based on image content.

B. Improved Downsampling Methods

Downsampling operations such as pooling and convolution with a stride larger than 1 are widely used in convolutional neural networks (CNNs) due to their capability to reduce computational overhead and memory usage while increasing the receptive field. However, downsampling methods can result in the loss of critical features, and several improved downsampling techniques have been proposed to mitigate such loss. The focus module in [21] and [27] rearranges input features to reduce spatial resolution while increasing channel dimensions, thus reducing computational overhead and reducing information loss. Conditional pooling [28] combines the strengths of max pooling and average pooling using a conditionally selective structure. Local importance-based pooling (LIP) [23] utilizes attention weights to average feature values within each pooling window, with uniform weights across elements in different windows. Neighbor feature attention-based pooling (NFP) [29] aggregates neighboring features to produce more discriminative downsampled representations guided by local context.

C. Summary of the Relationship Between CADM and Related Works

The proposed CADM can be considered both an advanced feature engineering-based method and an improved downsampling technique tailored for small object detection.

While most advanced feature engineering approaches aim to enhance small object detection, they often neglect the feature loss occurring during downsampling. In contrast, CADM incorporates a content-aware strategy to better preserve and aggregate small-object-related features during the downsampling process. Moreover, CADM differs from super-resolution-based methods [19], [20], which require high-resolution ground truth images—resources that are often unavailable in real-world remote sensing scenarios. Feature pyramid networks (FPN) [30] and path aggregation networks (PAN) [31] are widely adopted to improve detection by integrating multiscale features. CARAFE [26], a content-aware upsampling technique used in pyramid structures, does not rely on high-resolution supervision but still requires high-level feature maps with sufficient quality as input. Unfortunately, small-object-related features are frequently lost during downsampling, making it difficult for CARAFE to reconstruct them. In contrast, our proposed CADM only requires low-level feature maps, even original images as input, and then generates high-quality downsampled feature maps to improve small target detection performance.

Unlike existing improved downsampling techniques [23], [28], which are often optimized for general object detection or classification, CADM is specifically designed for small object detection. CADM generates spatially adaptive weights based on local content for each position in the feature map, enabling better retention of small-object-related information during downsampling. Compared with our earlier conference version [1], this work provides a more comprehensive introduction to the proposed method, includes extensive ablation studies, and offers broader comparisons with state-of-the-art methods. The experimental results validate the superior performance of our proposed CADDN.

III. Proposed Method

The overall architecture of the proposed CADDN is illustrated in Fig. 3. CADDN is composed of three main components: a parallel backbone, a fusion neck, and a detection head. In the parallel backbone stage, multiscale high-dimensional features are independently extracted from the input RGB and IR modalities. Our proposed CADM is integrated into the parallel backbone to perform adaptive downsampling based on feature content. In the fusion neck stage, the feature representations from both modalities are fused. Finally, the detection head predicts the locations and categories of objects from the fused feature representations. Next, the detailed implementation of the proposed CADDN detector, and CADM module therein is elaborated.

A. Detector Structure

1) Backbone: For multispectral input data, we adopt a parallel CNNs-based backbone, which is widely used in multispectral object detection tasks [9]. One of the primary advantages of this architecture is its capability to independently and effectively extract features from RGB and IR modalities. The parallel CNNs backbone applies a set of learnable convolutional filters to each modality, generating modality-specific feature maps that capture distinct patterns and structures from the respective inputs.

In our implementation, we employ cross-stage partial network (CSPNet) [32] as the backbone for both RGB and IR modalities. CSPNet incorporates convolutional layers, batch

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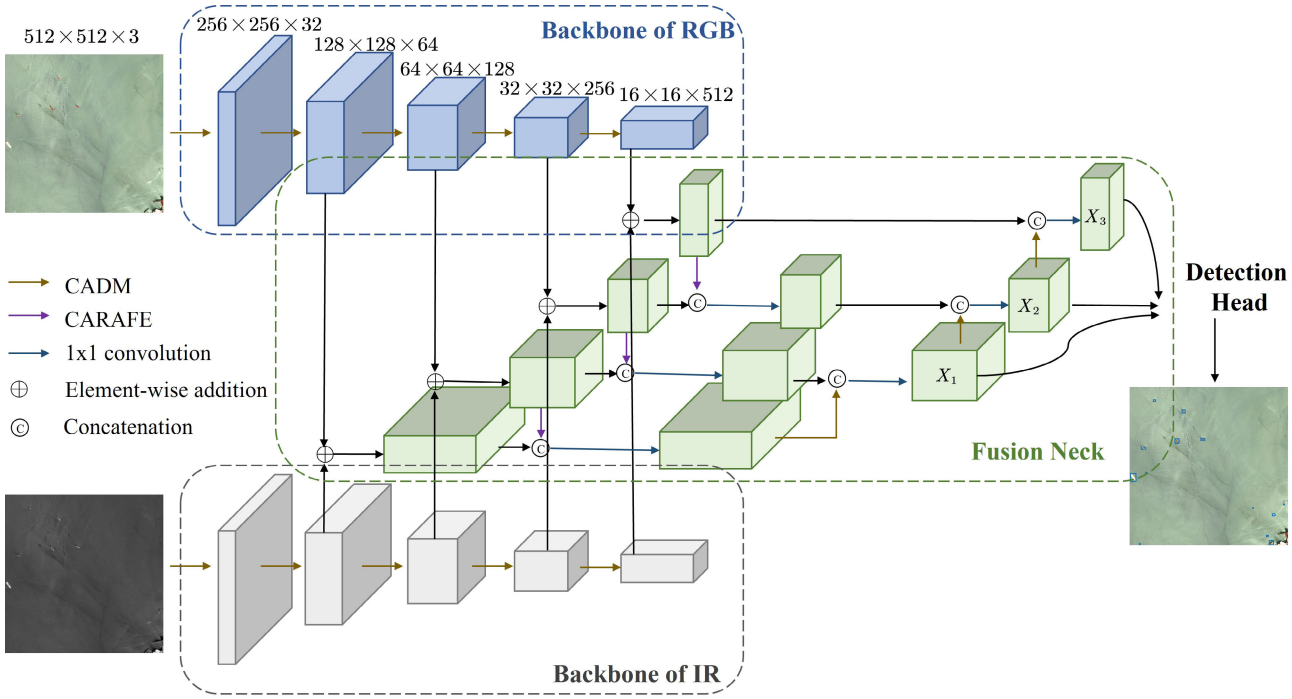


Fig. 3. Overall structure of the proposed CADDN.

normalization, and SiLU activation functions. During feature extraction, CSPNet performs downsampling at scales of 2, 4, 8, 16, and 32. Importantly, the proposed CADM is utilized as the primary downsampling operation in our backbone to better preserve critical information during the resolution reduction process.

2) Fusion Neck: In the fusion neck stage, the feature maps extracted from RGB and IR modalities are fused. We employ the additive fusion strategy [9] to combine the feature maps of the two modalities. Section IV further evaluates alternative fusion strategies. The fusion neck structure promotes multilevel fused feature maps containing information from both modalities, which corresponds to the green boxes in Fig. 3. Given the inherent discrepancy in feature distributions between RGB and IR modalities, we employ a composite structure that combines the FPN [30] and PAN [31] as fusion neck, and this structure has proven effective in enhancing detection performance and in constructing robust multiscale fusion representations. FPN propagates high-level semantic features to lower layers, thereby enriching the semantic capacity of low-level features. Conversely, PAN transmits low-level detailed features to higher layers, providing more accurate positional details of objects. As shown in Fig. 3, our fusion neck outputs multilevel feature maps $\{X_l, l \in 1, 2, 3\}$ that integrate both the positional and semantic information from RGB and IR modalities, where l is the index for the level of the feature map.

3) Detection Head: The detection head is responsible for predicting the class labels and locations of objects based on predefined anchor boxes. Each location (i, j) in the multilevel feature maps $\{X_l, l \in \{1, 2, 3\}\}$ corresponds to a specific grid cell in the original input image, determined by the receptive field. Each position on the feature map is preset with three anchors to accommodate objects with different aspect ratios. The detection head predicts the offset between the object center and the top-left corner of the corresponding grid cell $(\boldsymbol{c}_x, \boldsymbol{c}_y)$. Additionally, the detection head also predicts the width and height of the bounding box as ratios relative to the predefined anchor dimensions. As a result, the predicted object center $(\boldsymbol{b}_x, \boldsymbol{b}_y)$ and size $(\boldsymbol{b}_w, \boldsymbol{b}_h)$ are obtained. Simultaneously, the detection head predicts the class probability vector $\boldsymbol{c}_s$ and the objectness confidence score e_s. The predictions are generated from the multiscale feature maps using 1×1 convolution operations as follows:

$$(t_x, t_y, t_w, t_h, \boldsymbol{t}_c, t_e)_{i,j,l} = \mathrm{Conv}_{1\times1}(X_l(i, j)) \tag{1}$$

where (i, j) denotes the spatial location in the feature map. The vector $(t_x, t_y, t_w, t_h, \boldsymbol{t}_c, t_e)$ represents the raw output by the 1×1 convolution at each position, where (t_x, t_y, t_w, t_h) are the predicted offsets relative to the anchor boxes. To obtain the final bounding box predictions, the following transformations are applied:

$$\boldsymbol{b_x} = 2\sigma(\boldsymbol{t_x}) - 0.5 + \boldsymbol{c_x} \tag{2}$$

$$\boldsymbol{b_y} = 2\sigma(\boldsymbol{t_y}) - 0.5 + \boldsymbol{c_y} \tag{3}$$

$$\boldsymbol{b_w} = \boldsymbol{p_w}(2\sigma(\boldsymbol{t_w}))^2 \tag{4}$$

$$\boldsymbol{b_h} = \boldsymbol{p_h}(2\sigma(\boldsymbol{t_h}))^2 \tag{5}$$

where $\sigma(\cdot)$ denotes the sigmoid function, and $\boldsymbol{p}_w$ and $\boldsymbol{p}_h$ are the width and height of the corresponding anchor box, respectively. Finally, all predictions across different anchors and feature levels are aggregated, and redundant bounding boxes are eliminated using nonmaximum suppression (NMS) [33] to obtain the final object detection results.

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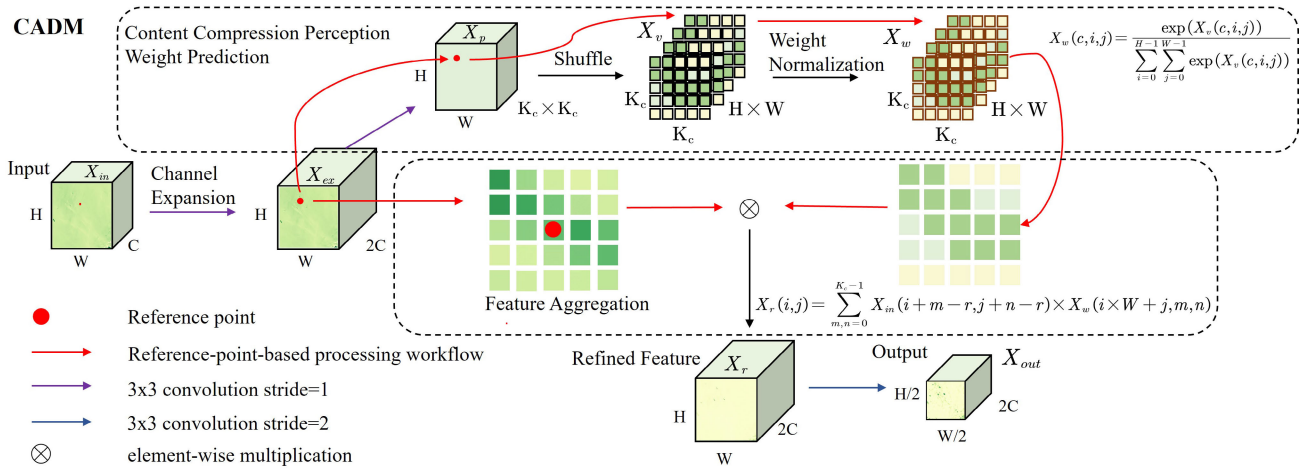


Fig. 4. Structure of the proposed CADM. The main operational process of CADM includes three operations: channel expansion, content compression perception weight prediction, and feature aggregation. This figure illustrates the processing flow of CADM for a specific position, using the red dot as the reference point (at a specific position), and the red arrows to represent the reference-point-based processing workflow.

B. Content-Aware Downsampling-Based Module

We replace the conventional downsampling operation in the CADDN backbone with the proposed CADM to mitigate the loss of object-relevant features and enhance small object representation. As illustrated in Fig. 4, CADM comprises three main steps: channel expansion, content-aware weight prediction, and feature aggregation. Let $X_{\text{in}} \in \mathbb{R}^{C\times H\times W}$ denote the input feature map, where C, H, and W are the number of channels, height, and width, respectively. CADM reduces the spatial resolution by a factor of two and doubles the number of channels, resulting in the output feature map $X_{\text{out}} \in \mathbb{R}^{2C\times(H/2)\times(W/2)}$. The input feature map first undergoes channel expansion to obtain channel-expanded features $X_{\text{ex}} \in \mathbb{R}^{2C\times H\times W}$. Content compression perception weight prediction is responsible for generating content aggregation weight X_w adaptively according to the content of X_{ex}. The feature aggregation utilizes the content aggregation weight X_w to aggregate spatial neighborhood small-object-related information, thereby generating the output feature map. Next, we introduce the steps of CADM in detail.

1) Channel Expansion: Conventional downsampling operations typically apply channel expansion to reduce information loss. Similarly, CADM adopts this approach by doubling the number of channels before downsampling, preserving key object-related details. The channel-expanded feature map $X_{\text{ex}} \in \mathbb{R}^{2C\times H\times W}$ is generated using a 3×3 convolution with a stride of 1

$$X_{\text{ex}} = \text{Conv}_{3\times 3,\text{stride}=1}(X_{\text{in}}) \tag{6}$$

where $\text{Conv}(\cdot)$ denotes a standard convolution operation with a kernel size of 3×3. This operation not only increases feature dimensions but also prepares the feature map for subsequent content-aware processing.

2) Content Compression Perception Weight Prediction: Content compression perception weight prediction is to perceive the content of the input feature map and estimate content aggregation weight for each position in the feature map. A 3×3 convolution with stride 1 is first applied to encode the channel-expanded feature X_{ex} into content compression perception code $X_p \in \mathbb{R}^{(K_c\times K_c)\times H\times W}$

$$X_p = \text{Conv}_{3\times 3,\text{stride}=1}(X_{\text{ex}}) \tag{7}$$

where K_c represents the preset content perception window size. Then, the channel dimension and the spatial dimension of the content compression perception code are reorganized through the shuffle operation as shown in Fig. 4. Through the shuffle operation, the content-aware kernel $X_v \in \mathbb{R}^{(H\times W)\times K_c\times K_c}$ is obtained

$$X_v(c,i,j) = X_p\left(i\times w + j, \left\lfloor \frac{c}{W}\right\rfloor, c \bmod K_c\right) \tag{8}$$

for all the spatial positions in the input feature map, the shuffle operation obtains $H\times W$ sets of content-aware kernels and each element in $X_v(c)$ represents the importance of each neighborhood element of position $(\lfloor c/W\rfloor, c \bmod W)$ in the input feature map X_{in}.

The content-aware kernels X_v are normalized in the spatial dimension by the softmax function before its application to the input feature map X_{in}, this ensures that the weighted result does not exhibit bias or scale differences compared with the original input feature map. Weight normalization can be expressed as follows:

$$X_w(c,i,j) = \frac{\exp(X_v(c,i,j))}{\sum_{i=0}^{H-1}\sum_{j=0}^{W-1}\exp(X_v(c,i,j))} \tag{9}$$

where X_w denotes the content aggregation weight. The content aggregation weights reflect the relative importance of each region in the feature map, enabling CADM to effectively aggregate information from neighboring pixels. This ensures the retention of small-object-related features that might otherwise be lost during conventional downsampling.

3) Feature Aggregation: The content aggregation weight X_w is used to aggregate the neighborhood features in X_{in}, thereby obtaining the refined feature X_r. Specifically, for a given position (i,j), we have

$$X_r(i,j) = \sum_{m,n=0}^{K_c-1} X_{\text{in}}(i+m-r, j+n-r) \times X_w(i\times W + j, m, n) \tag{10}$$

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where $r = \lfloor K_c/2 \rfloor$. In essence, our CADM aggregates the small-object-related neighborhood features within a $K_c \times K_c$ window centered on the reference point through content aggregation weight X_w to obtain the refined feature, as shown in Fig. 4. The refined feature contains stronger and more contrasting semantic information than the input feature map because small-object-related information in the local range can be more attended. Finally, the 3×3 convolution with a stride of 2 is applied to the refined feature X_r to obtain the downsampled output feature map X_{out} with the help of content awareness and neighborhood feature aggregation

$$X_{\text{out}} = \text{Conv}_{3\times3,\text{stride}=2}(X_r). \tag{11}$$

Compared with classic downsampling methods such as convolution with the stride of 2 and pooling, CADM adaptively adjusts the downsampling process based on content rather than fixed convolution parameters or artificial rules (such as preserving the maximum value in the local windows), thereby motivating the preservation of small-object-related discriminative features.

C. Loss Function and Training Strategy

Training CADM independently to perceive small object features is challenging due to the inherent ambiguity of small object boundaries and the coarse granularity of horizontal bounding box annotations commonly used in object detection. To address this, we integrate CADM into the object detection framework and adopt a joint training strategy. This approach enables CADM to learn content-aware capabilities guided by the supervisory signals from the detection task. In this way, CADM dynamically assigns greater weights to regions that are more relevant to small objects, thereby enhancing detection performance. The joint training paradigm ensures that the content awareness of CADM contributes meaningfully to the overall detection objective.

The design of the loss function is not the focus of this work, as CADM is compatible with existing deep-learning-based object detectors and can be integrated without altering their original loss functions. Therefore, we adopt the loss function proposed in [27], which includes three components: bounding box regression loss (L_{reg}), objectness confidence loss (L_{conf}), and classification loss (L_{clc}). The overall loss is formulated as follows:

$$L = \alpha_1 L_{\text{reg}} + \alpha_2 L_{\text{conf}} + \alpha_3 L_{\text{clc}} \tag{12}$$

where α_1, α_2, and α_3 are weighting coefficients that control the relative contribution of each loss term. These coefficients follow the settings recommended in [27].

The regression loss evaluates how accurately the predicted bounding boxes match the ground truth boxes. We employ the complete IoU (CIoU) loss [34], which considers overlap area, center point distance, and aspect ratio consistency between predicted and ground truth boxes. The regression loss is defined as follows:

$$L_{\text{reg}} = \sum_{i=1}^{n} \beta_i \cdot L_{\text{CIoU}}(b_i, \hat{b}_i) \tag{13}$$

where b_i and $\hat{b}_i$ represent the predicted and ground truth bounding boxes for sample i, respectively, and $L_{\text{CIoU}}(b_i, \hat{b}_i)$ is the CIoU loss between these boxes. The term β_i is a weighting factor used to balance the contributions from different feature map levels, and n is the total number of predicted bounding boxes.

The confidence loss measures the accuracy of the prediction regarding whether a bounding box contains an object. It is computed using binary cross-entropy (BCE) loss

$$L_{\text{conf}} = -\sum_{i=1}^{n} \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right] \tag{14}$$

where y_i is the ground truth label (1 if the ith bounding box contains an object, 0 otherwise), and p_i is the predicted confidence score for that bounding box. The summation is taken over all n predicted bounding boxes.

The classification loss evaluates how accurately the model assigns a predicted class label to each object detected in a bounding box. This is also computed using the BCE loss and is defined as follows:

$$L_{\text{clc}} = -\sum_{i=1}^{n} \sum_{c=1}^{C} \left[y_{i,c} \log(p_{i,c}) + (1 - y_{i,c}) \log(1 - p_{i,c}) \right] \tag{15}$$

where $y_{i,c}$ is the ground truth label for class c in the ith bounding box, $p_{i,c}$ is the predicted probability for class c, and C is the total number of classes. The loss is computed over all predicted bounding boxes and classes.

IV. Experimental Results and Analysis

In this section, we evaluate the performance of the proposed method on multispectral datasets in ship and vehicle detection scenarios. We first introduce the datasets used in the experiments, followed by the experimental settings including the experimental environment, hyperparameter settings, and evaluation criteria. Then, we present both qualitative and quantitative results to demonstrate the effectiveness of our approach. Finally, we compare our method with several state-of-the-art techniques to highlight the superior performance of CADDN.

A. Dataset Description

1) Multispectral Tiny Ship Dataset: We build a multispectral tiny ship dataset (MTSD) to evaluate our method. MTSD contains 2065 images with 4353 ship instance annotations. Fig. 1(a) illustrates representative detection scenes in MTSD. The dataset is split into 1445 training, 207 validation, and 413 testing image pairs. All images are captured by the Sentinel-2 multispectral satellite sensor with a spatial resolution of 10 m. The RGB modality is constructed from the R, G, and B channels, while the near-IR (NIR) modality uses the B8 band. All the images are processed to a pixel size of 512×512. The size distribution of ship targets is shown in Fig. 1(c), where the majority of objects are smaller than 32×32 pixels, making MTSD a challenging benchmark for small object detection.

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2) DroneVehicle: The DroneVehicle dataset [4] contains five object categories, i.e., car, truck, bus, van, and freight car. Fig. 1(b) presents sample scenes from this dataset. The DroneVehicle dataset includes three illumination conditions: dark night, night, and day, with a total of 389 779 and 428 086 annotated instances in the RGB and IR modalities, respectively. All images are captured by drones, resulting in 28 439 image pairs. The dataset is split into 17 990 training, 1469 validation, and 8980 testing pairs. Each image is 640 × 512 pixels in size. Following the setup in [35], we convert the original oriented bounding boxes (OBBs) to horizontal bounding boxes (HBBs) by computing their minimum enclosing rectangles. The object size distribution is shown in Fig. 1(d). The dataset includes a large number of small-scale vehicle objects, making it well-suited for evaluating the effectiveness of small object detection methods.

B. Experimental Settings

1) Experimental Environment: All the experiments are conducted in a consistent environment using Python 3.7.16 and PyTorch 1.8.0. The hardware includes an Intel[1] Xeon[1] Gold 6248R CPU and a Tesla V100-SXM2-32 GB GPU. The operating system is CentOS Linux 7 (Core), with CUDA Toolkit version 11.1.

2) Hyperparameter Settings: To ensure fair comparisons, we use consistent hyperparameters across our method, the baseline [9], and all compared methods [9], [14], [19], [21], [26], [27], [32], [36], [37]. The momentum of SGD is set to 0.98. The warm-up strategy is applied for the first three training epochs, and we use cosine annealing to adjust the learning rate cyclically between 0.01 and 0.001. The loss function employs weights of 0.05, 2.0, and 0.5 for regression, confidence, and classification losses, respectively. The weights assigned to the outputs from different feature map levels are 4.0, 1.0, and 0.4. The positive sample weighting coefficient is 2.0, and the weight decay is set to 0.0005. Without loss of generality, we set the content perception window size K_c in CADM to 5.

3) Evaluation Criteria: To evaluate the detection performance of methods, we select a set of performance metrics, including precision (P), recall (R), $F1$-score, as well as average precision at the given IoU threshold, specifically denoted as AP@0.5, and mean average precision, denoted as mAP@0.5. To define the above metrics, we first introduce the concepts of true positives (TP), false positives (FP), and false negatives (FN), where TP is the number of correct detections for each class, FP is the number of incorrect detections for each class, and FN is the number of undetected instances for each class. Precision (P) is defined as the ratio of TP to the sum of TP and FP, calculated separately for each class and then averaged. Mathematically, it is expressed as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{16}$$

Recall (R) quantifies the proportion of TP relative to the sum of TP and FN. This metric is also computed individually for each class before taking the average. The formal definition of recall is

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{17}$$

Given that precision and recall often exhibit a trade-off, the $F1$-score is employed as a harmonic mean of these two metrics. It is calculated as follows for each class:

$$F1 = 2 \times \frac{P \times R}{P + R}. \tag{18}$$

AP@0.5 is the area under the precision–recall curve for a particular class, evaluated at the IoU threshold of 0.5. It serves as a summary measure of the performance for that class, integrating both precision and recall

$$\text{AP@0.5} = \int_0^1 P(R) dR \tag{19}$$

mAP@0.5 is the mean of AP@0.5 calculated across all the classes and provides a singular as well as comprehensive metric that encapsulates the performance across multiple classes

$$\text{mAP@0.5} = \frac{1}{N} \sum_{i=1}^{N} \text{AP}_i\text{@0.5} \tag{20}$$

where i indexes the category, and N represents the total number of categories in the dataset. The aforementioned metrics provide a comprehensive and robust framework for evaluating object detection performance from multiple perspectives.

C. Analysis of CADM

In this section, we conduct a series of visualization experiments to intuitively illustrate how CADM addresses the challenge of small object detection in remote sensing images.

1) Visualization of Content Aggregation Weights: The core functionality of CADM lies in encoding the content information of the input feature map and aggregating the small-object-related feature through content compression perception and predicting the content aggregation weights. To provide an intuitive understanding of CADM, we visualize the content aggregation weights in Fig. 5. Specifically, we select multiple reference points (as defined in the legend of Fig. 4) and observe their corresponding aggregation weights. For visualization, a threshold of 0.05 is uniformly applied to filter out insignificant content perception weights, thereby highlighting the key regions CADM focuses on. The weights are then projected onto the input images based on their receptive fields.

Fig. 5(a) visualizes content aggregation weights for different locations on the same ship. As observed, CADM successfully focuses on object regions and tends to assign higher weights to areas where the object is present. When the reference point is located near the edge of the ship, CADM assigns higher importance to the outline and hull of the ship, while suppressing background responses. In contrast, Fig. 5(b) illustrates results in a background-only scene, where the aggregation weights are randomly dispersed with no discernible geometric pattern. This comparison underscores the capacity of CADM to concentrate on object-relevant regions in the presence of a target.

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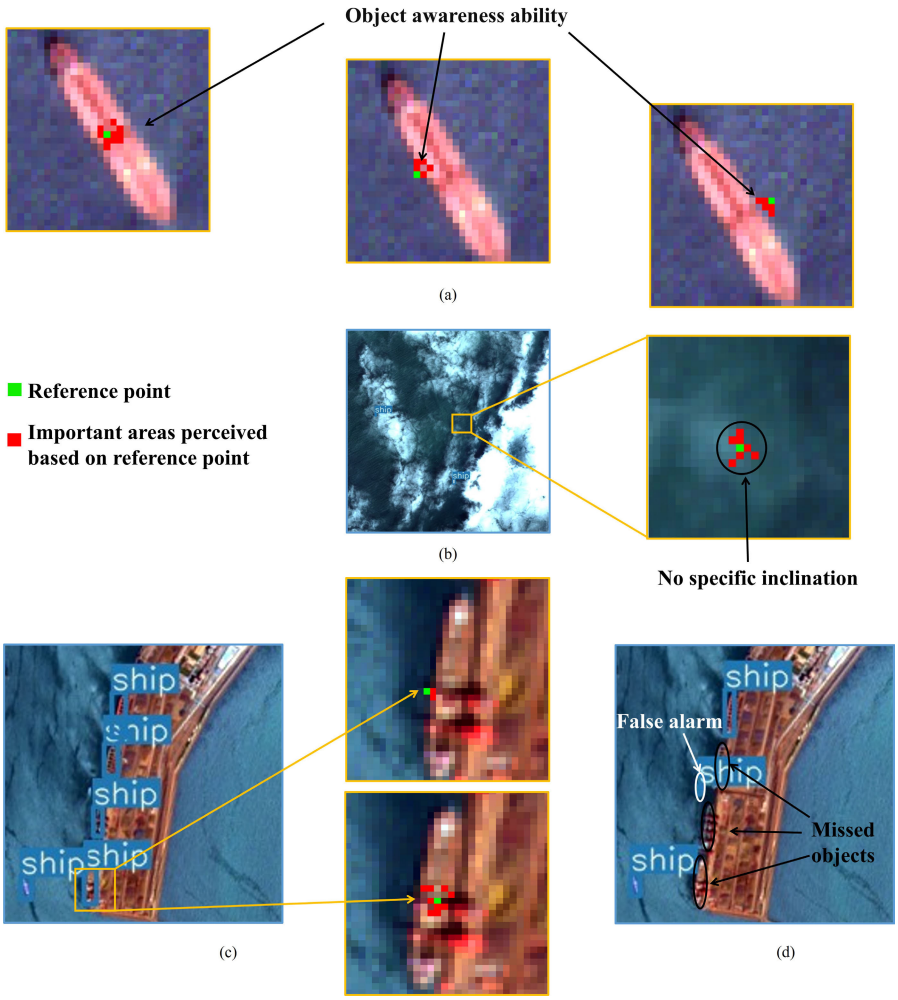


Fig. 5. Visualization of content aggregation weight. (a) Visualization of content aggregation weight with target region. (b) Visualization of content aggregation weight without target region. (c) Cases of successful detection by CADDN and the corresponding visualization results of content aggregation weights. (d) Cases of failed detection by the baseline. In (a)–(c), the green points represent reference points, and the red points surrounding the green dots correspond to values in the content aggregation weight of CADM exceeding the threshold value of 0.05. Red points represent important regions captured by CADM.

Fig. 5(c) and (d) show successful detection cases by CADDN and failed detection cases by the baseline under the same scene, respectively. This comparison intuitively highlights the advantage of integrating CADM. To further illustrate how CADM works for small object detection, Fig. 6 presents feature maps before and after downsampling for both CADDN and the baseline, i.e., the 3×3 convolution with a stride of 2. Fig. 6(a) displays the content aggregation weights, demonstrating the function of CADM. A comparison of Fig. 6(b) and (c) reveals that the contrast between the target and the background of the downsampled feature maps of CADDN is more pronounced than that in the baseline. To further validate that CADM prioritizes small object features rather than merely responding to high-frequency textures, we extend our experiments by incorporating nearshore scenarios in the last row of Fig. 6. In this setting, the reference point is chosen at the boundary between the shoreline, ship edge, and sea surface, a region with strong intensity variations. The results demonstrate that CADM concentrates on the object rather than the shoreline or sea surface. Besides, we use Grad-CAM [38] to visualize the attention heatmaps of our method and the baseline as shown in Fig. 7. The redder areas denote those that are more focused by the detector. Our proposed CADM successfully helps the detector to focus on the small object areas and allows the detector to better retain small-object-related features.

TABLE I
COMPARISON OF DETECTION EFFICIENCY AMONG DIFFERENT METHODS

Method	AP@0.5(%)	R(%)	P(%)	F1(%)	GFLOPs	Params(M)
Addfusion(s) [9]	56.9	52.2	72.0	60.5	**8.7**	**10.8**
Addfusion(m) [9]	59.1	55.7	68.7	61.5	26.8	31.9
Addfusion(l) [9]	59.5	55.1	75.4	63.7	60.7	70.3
CFT [21]	58.0	51.9	**76.8**	61.9	12.9	42.5
Biformer [37]	57.7	51.5	75.0	61.1	8.9	11.1
Ours	**62.2**	**58.1**	71.8	**64.2**	25.8	11.2

2) Computational Efficiency of CADM: Introducing CADM into the detector inevitably leads to more parameters and computational overhead. To fairly and comprehensively illustrate the superiority of CADM, we present an analysis in Table I. We compare the efficiency and performance of different methods. The results show that CADDN achieves a

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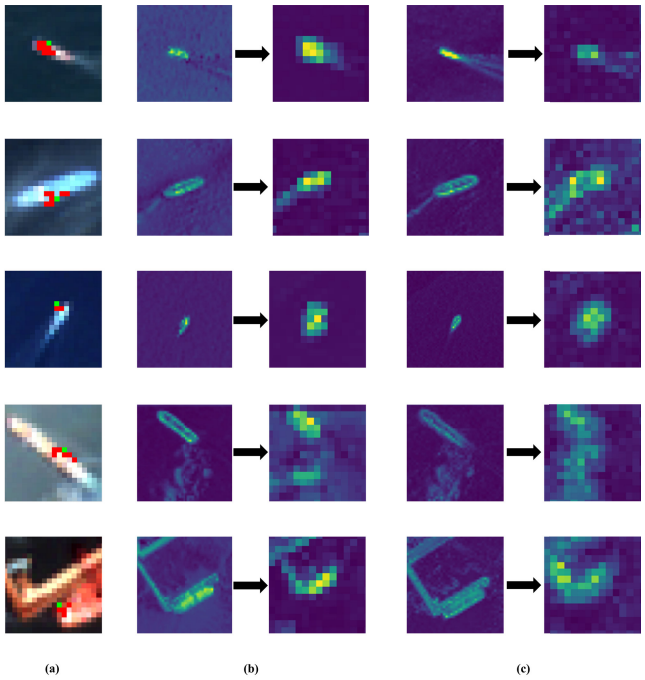


Fig. 6. Visual comparison of feature maps before and after downsampling. The spatial size of the feature map before downsampling is two times that of the original image, and the spatial size of the feature map after downsampling is eight times that of the original image. (a) Visualization of content aggregation weight with target region. (b) Visualization of feature maps before and after CADDN downsampling. (c) Visualization of feature maps before and after baseline downsampling, i.e., the 3×3 convolution with a stride of 2.

balanced trade-off: its efficiency ranks moderately among all compared methods, and it does not incur significant increases in computational complexity or parameter storage.

D. Ablation Study

In this section, we conduct a comprehensive series of ablation studies to evaluate the contribution and effectiveness of each component in our proposed method.

1) Analysis of Channel Expansion of CADM: Channel expansion increases the number of channels in the input feature map, enabling richer representations during downsampling, but may also introduce overfitting. Therefore, we experimentally investigate the relationship between the number of channels and detection performance. As shown in Table II, we compare four variants: reducing the channels to $0.5C$, keeping them unchanged at C, and expanding to $2C$ and $4C$, where C denotes the baseline channel number. Results show that reducing channels leads to significant performance degradation due to severe information loss. Keeping the number unchanged yields suboptimal results, while expanding to $2C$ achieves the best performance, validating the importance of channel expansion. However, further expansion to $4C$ causes a performance decline, likely due to overfitting. These results suggest that the effect of channel number on performance is nonmonotonic, and overly wide channels may introduce redundancy.

2) Analysis of CADM and Other Content-Aware Methods: CARAFE [26] is a representative content-aware upsampling method that inspires our work. However, in small object detection tasks, upsampling methods such as CARAFE often struggle to reconstruct high-resolution small object features from downsampled features, due to information degradation in earlier downsampling stages. We conduct this ablation study related to CARAFE to demonstrate the necessity and superiority of our proposed CADM. Table III reports the experimental results, where Cov and CADM represent the use of convolution with the stride of 2 and the use of CADM as the downsampling method, respectively. NI and CARAFE represent using nearest neighbor interpolation in [27] and CARAFE [26] as the upsampling method, respectively. First, introducing CADM alone improves performance over the baseline, confirming the effectiveness of CADM. Second,

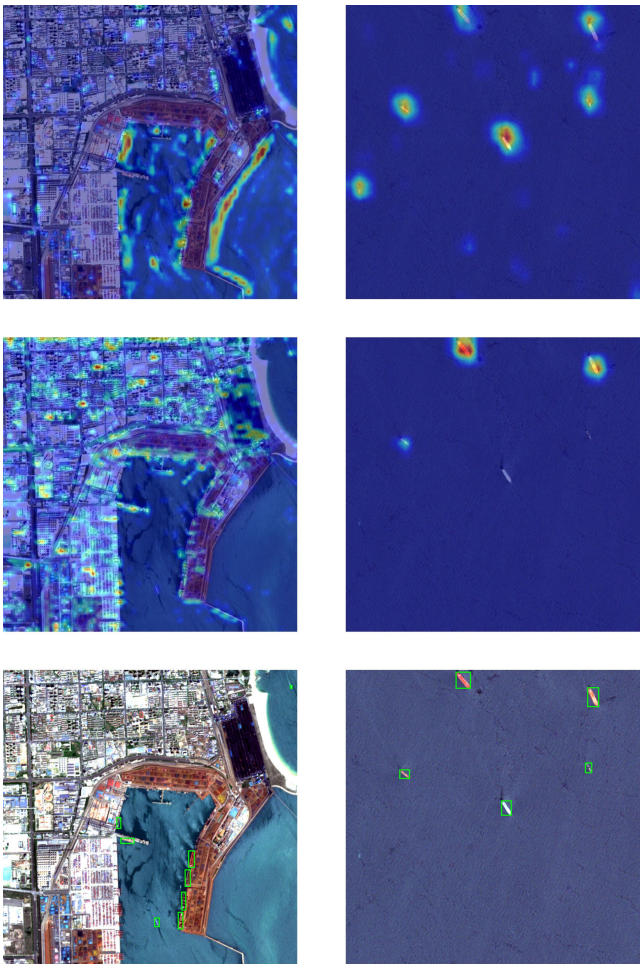


Fig. 7. Attention heatmaps for detectors visualized by Grad-CAM [38]. The first, second, and third lines are the result of CADDN, the result of the baseline, and the third line is the ground truth, respectively.

TABLE II
EXPERIMENT ON THE NUMBER OF CHANNELS IN CHANNEL EXPANSION OF CADM

Channels	AP@0.5(%)	R(%)	P(%)	F1(%)
0.5C	56.9	52.2	70.0	59.8
C	60.5	55.2	**75.4**	63.6
2C	**62.2**	**58.1**	71.8	**64.2**
4C	57.1	54.6	72.8	62.4

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TABLE III
EXPERIMENTS ON COMPARING CADM WITH OTHER ATTENTION METHODS

Backbone	Downsampling	Upsampling	AP@0.5(%)	R(%)	P(%)	F1(%)
CSPNet	Cov	NI [27]	58.8	52.9	75.9	62.3
CSPNet	CADM	NI [27]	60.0	55.0	75.8	63.7
CSPNet	Cov	CARAFE [26]	58.5	53.5	**76.9**	63.1
CSPNet	CADM	CARAFE [26]	**62.2**	**58.1**	71.8	**64.2**
CSPNet	Sp-Att [39]	CARAFE [26]	59.3	54.8	72.4	62.4
CSPNet	Self-Att [40]	CARAFE [26]	59.1	54.5	75.1	63.2

using CARAFE alone shows limited improvement, reflecting that critical feature loss often occurs during the downsampling process in the small object detection task, and this loss prevents CARAFE from effectively learning and recovering features. Finally, we use CARAFE to replace the upsampling method in FPN as well as employ CADM to replace the downsampling method in our detector, conducting experiments under all other equal conditions. The combination of CADM and CARAFE yields better performance compared to only using CARAFE or only using CADM. These experimental results validate the rationale behind CADM, i.e., the loss of small-object-related features will lead to a series of training challenges, rendering CARAFE ineffective. However, the introduction of CADM ensures the preservation and aggregation of important features, providing a better foundation for the effectiveness of CARAFE.

The attention mechanisms [39], [40] are representative content-aware methods, as they adaptively adjust feature representations based on the input content. However, existing attention mechanisms such as spatial-attention (Sp-Att) [39] and self-attention (Self-Att) [40] are not explicitly designed for downsampling in small object detection tasks for remote sensing images, leading to suboptimal performance. To investigate this issue, we replace the content compression perception weight prediction layer in CADM with Sp-Att [39] and Self-Att [40] to construct alternative downsampling methods. The comparative results in Table III demonstrate the superior performance of our method. Unlike Sp-Att, which globally assigns an importance score to each pixel without considering its relative significance in different contexts, our method dynamically predicts the importance of each pixel in relation to its surrounding neighborhood. This capability is particularly crucial in remote sensing small object detection, where features such as object boundaries exhibit different significance depending on the reference region. For example, the contour of a ship is important relative to its hull but less significant relative to the nearby shoreline. Sp-Att [39], which applies a uniform importance score, may lead to confusion between objects and backgrounds. In contrast, CADM learns fine-grained spatial dependency by computing per-pixel relative importance within local regions, generating more detailed feature maps that are beneficial for small object detection in remote sensing images. Moreover, Self-Att [40] captures long-range dependency by modeling global correlations, but this property is not well-aligned with the requirements of downsampling for small object detection. Downsampling relies primarily on local information, and incorporating long-range dependency may introduce unnecessary complexity, leading to parameter redundancy and increased training difficulty.

3) Analysis of Using Low-Level Feature Maps: Existing object detectors usually use feature maps downsampled by the factor of 8, 16, and 32 to form multilevel feature maps [9], [19], [21], [27], and the feature maps downsampled by the factor of 2, 4, 8, 16, and 32 are denoted by $P1$, $P2$, $P3$, $P4$, and $P5$, respectively. However, due to the inevitable loss of small-object-related features during the downsampling process, high-resolution low-level feature maps may play an important role in detection. Therefore, we attempt to introduce the feature maps downsampled by the factor of 4 (which is denoted as $P2$) into the FPN to construct an effective and reasonable small object detection method. In Table IV, we do not use any content-aware methods to improve the upsampling or downsampling operators but only study the effect of using low-level feature maps. The comparison from Table IV reveals that the introduction of $P2$ significantly improves detection performance. However, as shown in Table IV, introducing a feature map with downsampling by a factor of 2 ($P1$) does not effectively improve the performance of the detector. This is because $P1$ contains not only detailed features of the target but also other interfering factors such as background noise. Considering the observations in Table IV, we choose to introduce $P2$ in FPN for the next experiments.

4) Analysis of Using High-Level Feature Maps: The process of detecting small objects may result in the loss of important features due to the downsampling of feature maps through the deeper layers of a neural network. However, the preservation and utilization of high-level features remain imperative for a number of reasons, particularly in the context of small object detection.

1) The presence of high-level features affords a more expansive receptive field, thereby enabling the model to consider a greater range of structural relationships within the image. While low-level features are adept at capturing fine details such as edges and textures, the identification of small objects often necessitates a more expansive context.
2) While low-level features can retain object details, they frequently lack the capacity to provide sufficient distinction when confronted with cluttered or noisy backgrounds. Conversely, high-level features offer more abstract and semantically rich representations, thereby rendering them more effective in distinguishing objects from their surroundings in complex scenes.
3) The integration of both low-level and high-level features enhances the detection of small objects. Low-level features provide detail, whereas high-level features offer contextual and semantic information. The integration of features from multiple levels enables models to leverage fine-grained information while also benefiting from the broader context captured by deeper layers. Modern neural network architectures employ mechanisms such as FPNs to effectively fuse low- and high-level features. This ensures the preservation of fine details while simultaneously leveraging high-level abstractions,

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TABLE IV
EFFECTIVENESS EXPERIMENT VALIDATION OF INTRODUCING LOW-LEVEL FEATURE MAPS

Backbone	P1	P2	P3	P4	P5	AP@0.5(%)	R(%)	P(%)	F1(%)	GFLOPs	Params(M)
CSPNet			✓	✓	✓	56.9	52.2	72.0	60.5	**8.7**	**10.8**
CSPNet		✓	✓	✓	✓	**58.3**	54.9	**73.7**	**62.9**	10.2	10.9
CSPNet	✓	✓	✓	✓	✓	58.1	**56.9**	69.6	62.6	11.0	10.9

TABLE V
EFFECTIVENESS EXPERIMENT VALIDATION OF USING HIGH-LEVEL FEATURE MAPS

Backbone	P1	P2	P3	P4	P5	AP@0.5(%)	R(%)	P(%)	F1(%)
CSPNet	✓	✓	✓	✓	✓	**62.2**	**58.1**	71.8	**64.2**
CSPNet	✓	✓	✓	✓		61.0	53.2	**73.4**	61.7
CSPNet	✓	✓	✓			59.6	55.2	64.5	59.5

TABLE VI
EXPERIMENTAL VALIDATION OF THE OPTIMAL LEVEL USING CADM

Backbone	P0→P1	P1→P2	P2→P3	P3→P4	P4→P5	AP@0.5(%)	R(%)	P(%)	F1(%)
CSPNet	CADM	CADM	CADM	CADM	CADM	60.9	54.9	**74.7**	63.3
CSPNet	Focus	CADM	CADM	CADM	CADM	60.4	55.5	67.2	60.8
CSPNet	CADM	CADM	CADM	Cov	Cov	61.0	56.3	74.0	63.9
CSPNet	CADM	Cov	Cov	CADM	CADM	60.1	55.9	69.3	61.9
CSPNet	CADM	CADM	Cov	Cov	Cov	**62.2**	**58.1**	71.8	**64.2**

thereby enhancing the capacity to detect small objects with better precision.

We perform a series of experiments on our CADDN model to validate the necessity of using high-level features. For a fair comparison, we consistently use CADM to improve the downsampling process and use CARAFE [26] to improve the upsampling process. The results are summarized in Table V. As shown, the removal of high-level features ($P4$ and $P5$) leads to a consistent decrease in performance. This confirms that high-level features contribute significantly to the performance of small object detection.

5) Analysis of Using CADM at Different Levels: The feature maps of CADDN exhibit characteristics of gradually decreasing resolution and increasing semantic information; intuitively, we investigate whether CADM is universally applicable to all the downsampling positions. We set up to employ either CADM or convolution with the stride of 2 as the downsampling method between different levels of feature maps. Both CADM and convolution with the stride of 2 can achieve the downsampling function, but for high-quality low-level feature maps, CADM can obtain better downsampling feature maps through content awareness. Table VI presents our experimental results, where Cov, Focus, and CADM represent the use of convolution with the stride of 2, the use of the Focus module in [27], and the use of CADM as the downsampling method, respectively. In Table VI, better performance is achieved by using CADM only on low-level feature maps. Considering that low-level feature maps contain detailed object textures, and high-level feature maps, which are downsampled from low-level features by CADM, mainly represent semantic features of objects by high-intensity brightness, we can speculate that the success of CADM is based on learning and perceiving the geometric information of objects in the low-level feature maps. CADM uses the low-level feature map to preserve and aggregate important features, ensuring that the small-object-related features do not suffer significant loss and have a stronger contrast with the background after downsampling. However, since the detailed texture information has inevitably become blurred for high-level feature maps, CADM is not necessary here. As introduced in Section II-B, the baseline [21] not only uses convolution with the step greater than 1 as the downsampling method but also uses the focus module as the downsampling method in the input stage of the backbone. We replace the focus module with CADM to illustrate the advantages of CADM. As shown in Table VI, by comparing whether to choose focus replacement by CADM (i.e., whether to use CADM at $P0 \rightarrow P1$), we can see that the performance of using CADM is significantly better than using the focus module.

6) Analysis of Using Different Fusion Strategies: To address the challenge of integrating RGB and IR modalities for improved detection performance, we evaluate three fusion strategies: our used parallel backbone, the early fusion method using addition, and the early fusion method using concatenation. The early fusion methods use a unified backbone to process both RGB and IR channels, either by performing pixel-level addition (Add) or by concatenating the two inputs into a six-channel input (Cat). In contrast, the parallel backbone extracts features from the two modalities separately. We chose the parallel backbone for the following two reasons.

1) Extracting modality-specific features is a challenging task due to the inherent differences between the RGB and IR channels. While RGB channels are adept at capturing detailed color and texture information, IR channels excel in low-light environments. Attempting to process both modalities using a single network results

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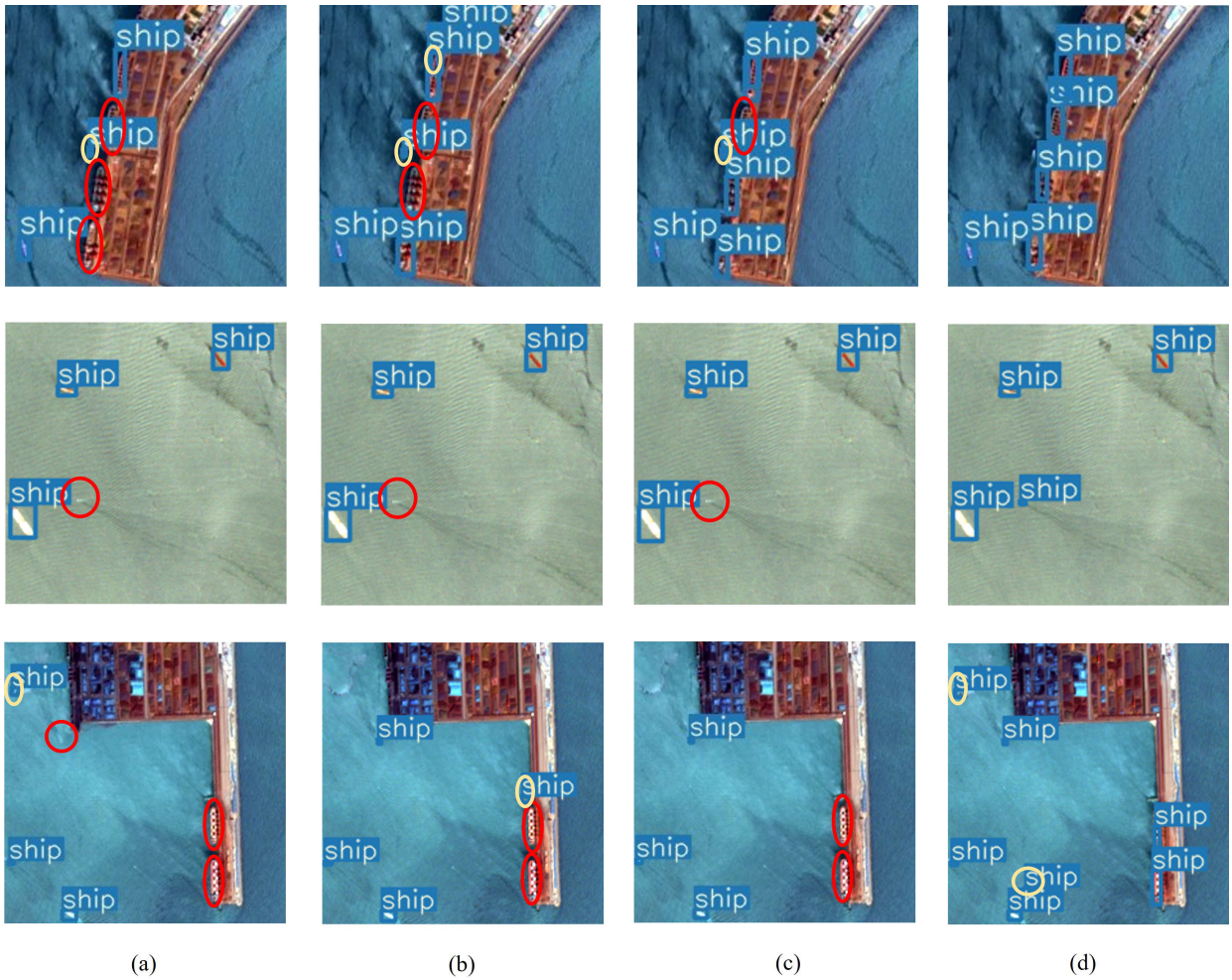


Fig. 8. Visual results of object detection using different methods on MTSD. The red cycles represent missed detections, the yellow ones denote the false alarm, and the detection results that are not circled are the correct results. (a)–(d) Results of add-fusion-based multispectral detector [9], CFT [21], DRENet [19], and our proposed CADDN, respectively.

TABLE VII
COMPARISON OF DIFFERENT FUSION STRATEGIES

Backbone	AP@0.5(%)	Recall (%)	Precision (%)	F1 (%)
Parallel Backbone	**62.2**	**58.1**	**71.8**	**64.2**
Early Fusion (Add)	61.0	55.7	70.0	62.0
Early Fusion (Cat)	61.7	57.3	71.6	63.7

in the loss of their unique characteristics, as they are fundamentally different. The use of a parallel backbone allows the model to independently extract both common and modality-specific features, enabling a flexible and effective exploitation of the strengths of each modality.

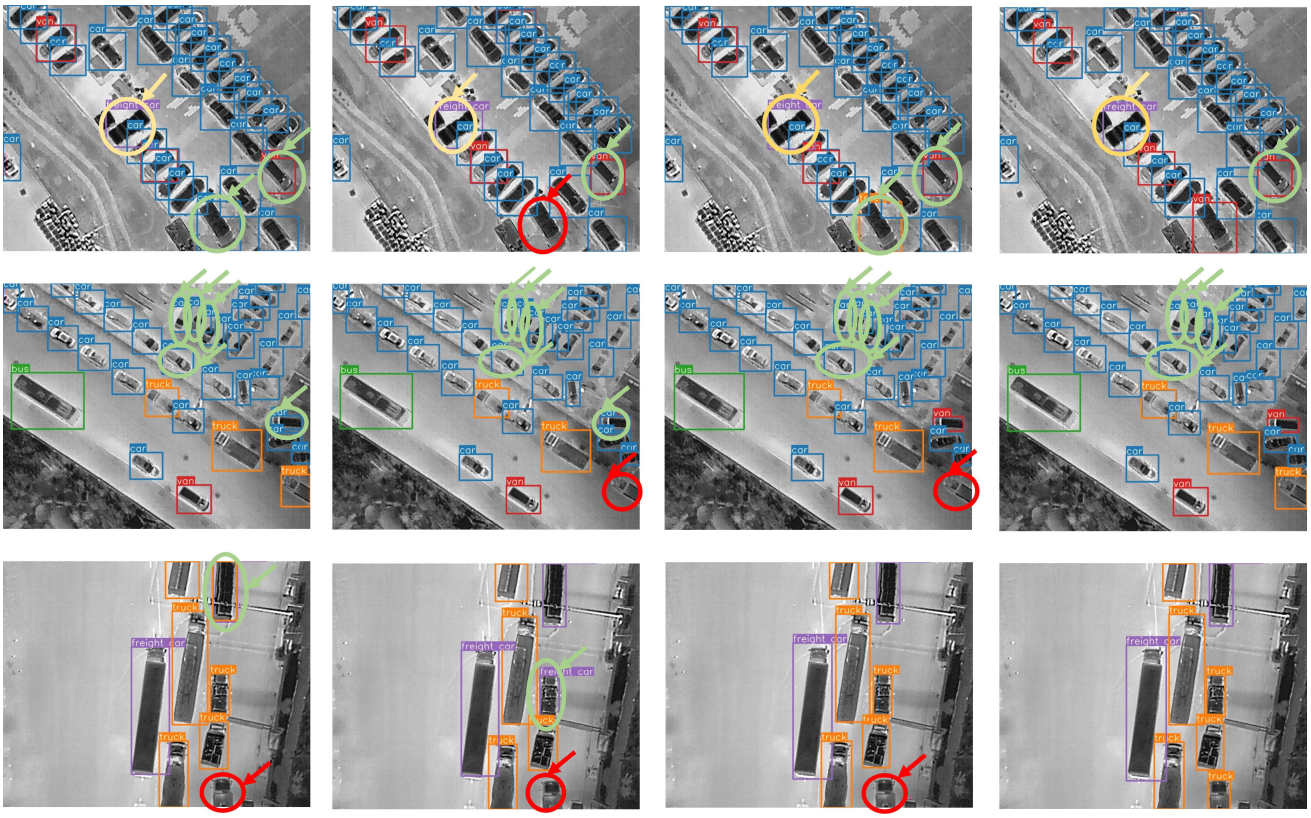
2) In early fusion methods, the dominance of a single modality can result in the overshadowing of modality-specific information. The parallel backbone circumvents this issue by treating the modalities separately, thereby preserving important features for detection.

In addition, we conduct a comparative experiment to justify the use of a parallel backbone. The results are presented in Table VII. The parallel backbone achieves the best performance compared to the early fusion methods. This confirms that treating RGB and IR channels separately, rather than using a backbone network to uniformly process RGB and IR modalities, leads to superior detection performance.

E. Comparison With State-of-the-Art Methods

We thoroughly validate the superiority of our method using MTSD and DroneVehicle dataset [4]. Under identical hyperparameters and experimental environment settings, we reproduce a series of remarkable works: add-fusion-based multispectral detector [9], you only look once version 5 (YOLOv5) [27], attentional scale sequence fusion YOLO (ASF-YOLO) [36], cat fusion-based multispectral detector [9], input fusion-based multispectral detector [9], DRENet [19], cross-modality fusion transformer (CFT) [21], CARAFE [26], NWD [14], and bilevel routing attention transformer (Biformer) [37]. Among them, ASF-YOLO [36], DRENet [19], NWD [14], and Biformer [37] are advanced small object detection methods, while add-fusion-based multispectral detector [9], cat fusion-based multispectral detector [9], input fusion-based multispectral detector [9], and CFT [21] are popular multispectral detector, CARAFE [26] is a kind of content-aware upsampling work, and YOLOv5 [27] is a widely used general

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(a) (b) (c) (d)

Fig. 9. Visual results of object detection using different methods on DroneVehicle. The red cycles represent missed detections, the yellow ones denote the false alarm, the green circles represent the wrong classification, and the detection results that are not circled are the correct results. (a)–(d) Results of add-fusion-based multispectral detector [9], CFT [21], DRENet [19], and our proposed CADDN, respectively.

TABLE VIII
QUANTITATIVE COMPARISON BETWEEN THE PROPOSED METHOD AND OTHER METHODS ON MTSD

Method	Backbone	Modality	AP@0.5(%)	R(%)	P(%)	F1(%)
Addfusion [9]	CSPNet	RGB+IR	56.9	52.2	72.0	60.5
YOLOv5 [27]	CSPNet	RGB	47.9	46.9	63.1	53.8
ASF-YOLO [36]	CSPNet	RGB	53.8	48.9	72.3	58.3
Catfusion [9]	CSPNet	RGB+IR	56.9	56.7	67.0	61.4
Inputfusion [9]	CSPNet	RGB+IR	57.0	56.2	70.3	62.5
DRENet [19]	CSPNet+CRMA	RGB	52.0	**58.8**	35.6	44.3
CFT [21]	CSPNet	RGB+IR	58.0	51.9	76.8	61.9
CARAFE [26]	CSPNet	RGB+IR	57.8	54.4	71.8	61.9
NWD [14]	CSPNet	RGB+IR	43.3	39.7	65.2	49.4
Biformer [37]	CSPNet	RGB+IR	57.7	51.5	75.0	61.1
Ours	CSPNet	RGB	61.5	54.8	**77.1**	64.1
Ours	CSPNet	RGB+IR	**62.2**	58.1	71.8	**64.2**

object detector. To ensure a fair comparison, we adopt almost identical backbone network configurations, concentrating on contrasting the impact of each algorithm on detection performance. When comparing our method with methods that use single modality data [19], [27], [36], we change the parallel backbone of CADDN to a single backbone architecture.

Tables VIII and IX present experimental results on MTSD and DroneVehicle, respectively. Given the detection task of the DroneVehicle dataset with five categories, i.e., car, truck, bus, van, and freight car, we provide AP@0.5 and mAP@0.5 for each category. On the MTSD dataset, our approach achieves the AP@0.5 of 62.2%, marking a 4.2% improvement with the nearest competitor [21]. In addition, our method outperforms the baseline approach add-fusion-based multispectral detector [9] by a 5.3% AP@0.5 margin. On the DroneVehicle dataset, our approach achieves the mAP@0.5 of 82.6%, registering a 4.0% improvement compared with the closest competitor [9] and surpasses the baseline method add-fusion-based multispectral detector [9] by a 4.5% mAP@0.5. Furthermore, the comparison between single-modal and multispectral detection experiments in Tables VIII and IX demonstrates the significance of utilizing both RGB and IR data. Although the precision of the CADDN model using both RGB and IR data is slightly lower than that of the RGB-only

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TABLE IX
QUANTITATIVE COMPARISON BETWEEN THE PROPOSED METHOD AND OTHER METHODS ON DRONEVEHICLE

Method	Backbone	Modality	mAP@0.5(%)	R(%)	P(%)	F1(%)	Car(%)	Truck(%)	Bus(%)	Van(%)	Freight Car(%)
Addfusion [9]	CSPNet	RGB+IR	78.1	73.5	73.2	73.3	98.0	79.6	94.8	54.8	63.4
YOLOv5 [27]	CSPNet	IR	73.5	68.9	73.0	70.9	97.6	72.7	91.4	46.6	59.0
ASF-YOLO [36]	CSPNet	IR	75.6	69.1	79.7	74.0	93.0	72.3	95.0	60.3	57.7
Catfusion [9]	CSPNet	RGB+IR	77.4	73.8	74.3	74.0	98.0	79.3	94.8	52.6	62.3
Inputfusion [9]	CSPNet	RGB+IR	78.6	75.5	73.8	74.6	98.0	79.4	95.1	56.3	64.3
DRENet [19]	CSPNet+CRMA	IR	77.5	**87.9**	48.7	62.7	97.2	75.3	94.3	57.5	63.3
CFT [21]	CSPNet	RGB+IR	78.2	74.7	74.4	74.5	98.0	79.9	95.1	54.5	63.3
CARAFE [26]	CSPNet	RGB+IR	78.1	75.4	73.8	74.6	98.0	80.3	94.7	55.0	62.3
NWD [14]	CSPNet	RGB+IR	74.5	73.1	68.7	70.8	95.1	74.7	93.1	50.5	59.1
Biformer [37]	CSPNet	RGB+IR	77.3	75.6	71.5	73.5	98.0	80.3	95.1	52.2	61.0
Ours	CSPNet	IR	77.8	73.0	73.3	73.1	97.9	80.1	94.7	52.5	63.9
Ours	CSPNet	RGB+IR	**82.6**	76.4	**81.0**	**78.6**	**98.2**	**84.6**	**96.0**	**63.7**	**70.3**

data on the MTSD dataset, this does not indicate inferior performance. Precision only reflects the accuracy of positive predictions but does not account for the capability of the model to detect all relevant targets. Both AP@0.5 and $F1$-score are more comprehensive evaluation metrics and are consistently higher for the multispectral model than the RGB-only model, highlighting the effectiveness of incorporating IR information through multispectral fusion.

Figs. 8 and 9 display the representative detection results of the add-fusion-based multispectral detector [9], CFT [21], DRENet [19], and our method in challenging detection scenarios on the MTSD and DroneVehicle datasets. Compared to these methods, our proposed CADDN shows a significant improvement in small-object recall and reduction of false alarms. The add-fusion-based multispectral detector struggles with small object detection due to its traditional downsampling method and simple fusion strategy, which cannot effectively distinguish small object features from background noise. CFT, while leveraging global context through self-attention, lacks specialized mechanisms for enhancing small object details, leading to reduced performance for smaller targets. DRENet focuses on suppressing background interference but is limited in its ability to preserve fine-grained small object features. In contrast, our CADDN utilizes the proposed CADM to adaptively preserve and enhance small-object-related features during downsampling, resulting in more accurate detection results.

V. CONCLUSION

In this article, we propose CADDN for small object detection in multispectral remote sensing images, where a novel content-aware downsampling-based module, i.e., CADM, is designed to replace the current standard downsampling approaches. By encoding the content of the input feature map, and predicting the importance of neighborhood information at each location as the content aggregation weight, CADM effectively perceives and aggregates small-object-related features based on the content during the downsampling process. Experiments demonstrate the superiority of our proposed method on both MTSD and DroneVehicle datasets. In our view, a more lightweight version of CADM is a valuable research direction in the future.

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